

An aerial night view of a city, likely Shanghai, with a network overlay of glowing nodes and connecting lines. The city lights are visible, and the network lines are white and curved, connecting various points across the city. The sky is a deep blue, and the city lights are warm yellow and orange.

Wireless Networks

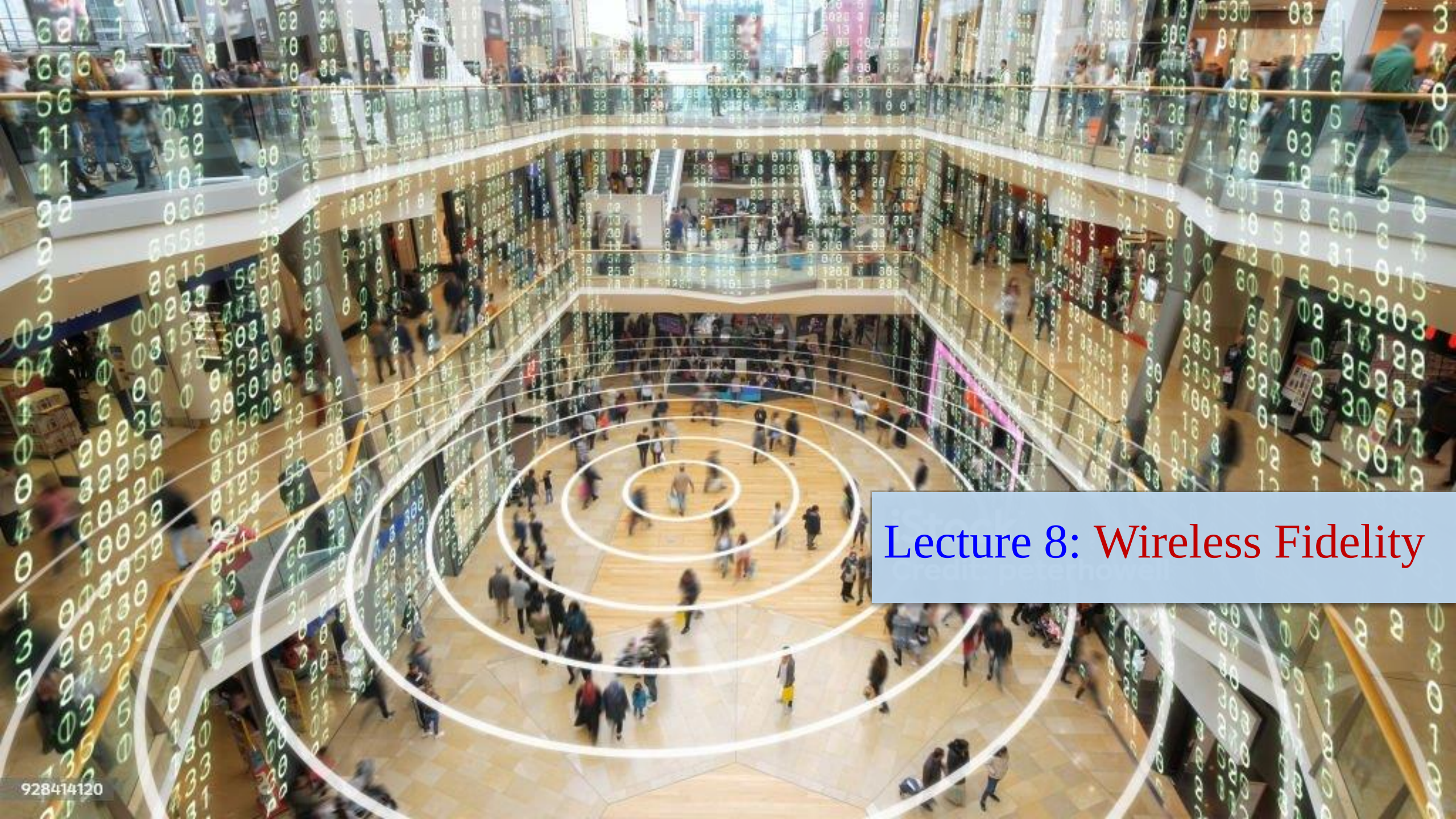
Prof. Dr. Hassan Al-Mahdi

Head of the Computer Science Department

Faculty of Computer and Informatics

Suez Canal University

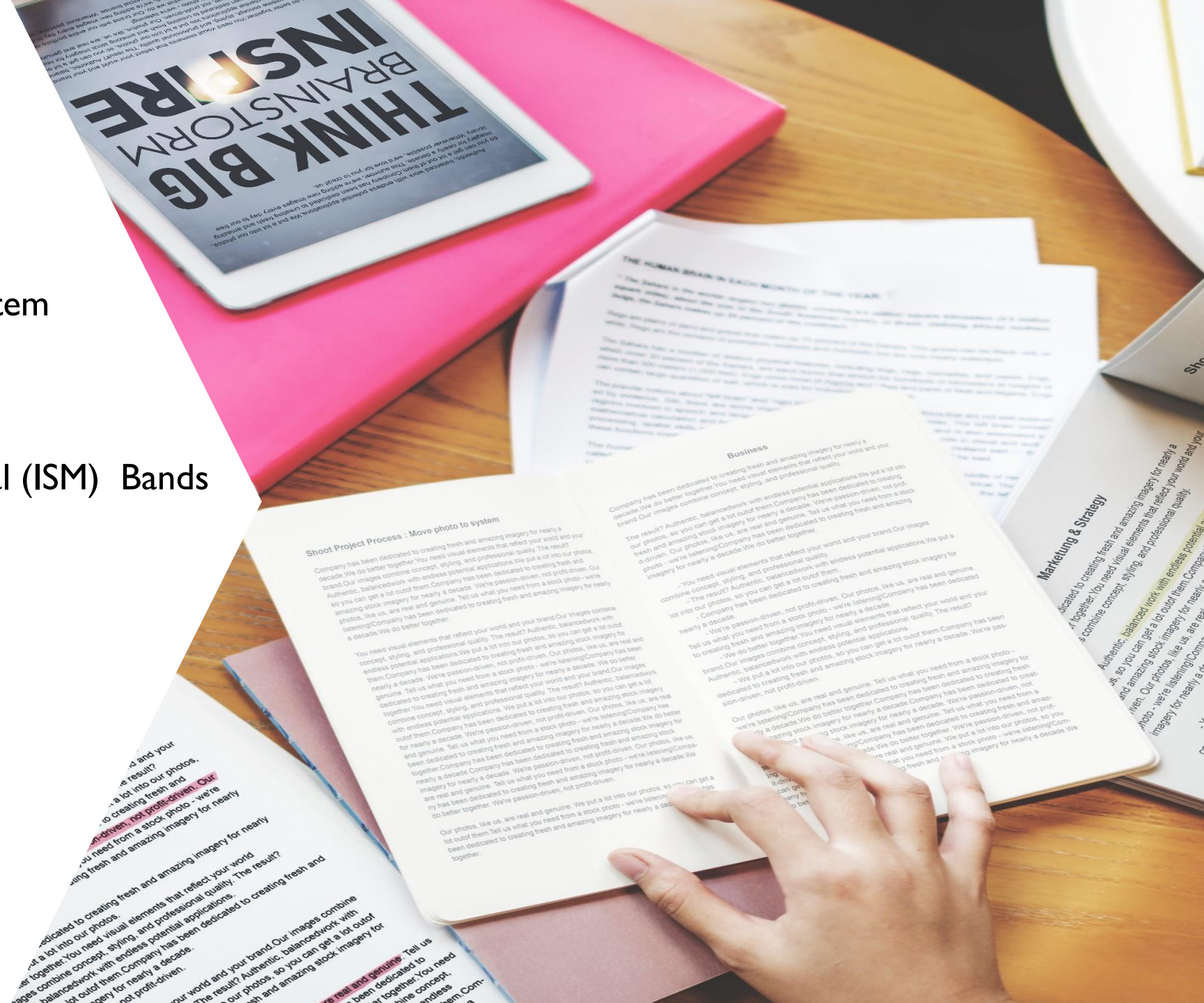
2024-2025



Lecture 8: Wireless Fidelity

Contents

- ✓ Wireless Fidelity
- ✓ WiFi vs IEEE 802.11
- ✓ IEEE Standards Numbering System
- ✓ IEEE 802.11 Features
- ✓ Unlicensed Spectrums
- ✓ Industrial, Scientific, and Medical (ISM) Bands
- ✓ IEEE channels
- ✓ Physical Layers
- ✓ Hidden Node Problem



Wireless Fidelity

☑ **WiFi** stands for **W**ireless **F**idelity:

- ✓ Available in **indoor** and **outdoor** public places such as:
 - *Homes and airports*
 - *Shopping malls and parks*
 - *University campuses.*
- ✓ Popular option for internet connectivity.
- ✓ **Evolved significantly** over the last decade, which resulted in increasing its **complexity**.

WiFi versus IEEE 802.11

☑ IEEE 802.11:

- The Institute of Electrical and Electronics Engineers (IEEE)
- Designed for or wireless local area networks (WLANs).

☑ It specifies the protocols for how devices communicate wirelessly:

- Physical layer (radio frequencies, modulation)
- Data link layer (MAC addressing, access control).

☑ IEEE 802.11 includes versions each improving speed, range, or efficiency :

- 802.11a, 802.11b, 802.11g, 802.11n and 802.11ac
- 802.11ax for WiFi 6.

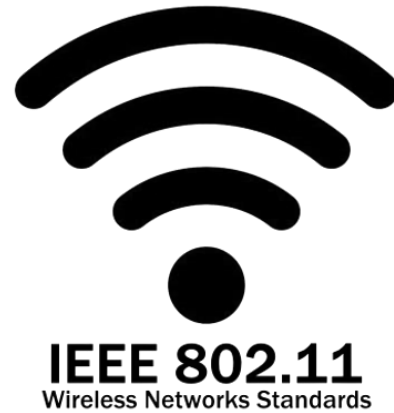
WiFi vs IEEE 802.11

☑ **IEEE 802.11** and **Wi-Fi** refer to **wireless networks (WLANs)** that allow devices to connect without cables.

☑ **IEEE 802.11** is a **technical standard** developed by the **IEEE** that defines how wireless networks work.

☑ **IEEE 802.11** has many options:

- 1) It has features and options that companies (vendors) can choose to implement in their devices.
- 2) Different routers, laptops, and phones may use different settings, even if they all follow IEEE 802.11.
- 3) Different vendors implement IEEE 802.11 features differently, causing compatibility issues.
- 4) For example, a router from Brand A may not fully support a laptop from Brand B, leading to slow speeds or dropped connections



WiFi vs IEEE 802.11

- ☑ **Wi-Fi** is a **certification** given by the **Wi-Fi Alliance** to ensure devices from different brands work together.



- ☑ **Wi-Fi Alliance Ensures Compatibility**

- ✓ It **selects a set of IEEE 802.11 features** that must be supported for a device to be **Wi-Fi certified**.
- ✓ This ensures that Wi-Fi-certified devices (like phones, laptops, and routers) from different brands **work together smoothly**.
- ✓ The Wi-Fi logo ensures product compatibility

<https://www.ieee802.org/11/>

Standards Numbering System

- ✓ **IEEE** has a unique numbering system for its IEEE wireless LAN standards.
- ✓ In the early **1980s**, the group **802** was formed to define various network functionalities.
- ✓ 802 defined the following standards:
 - **IEEE 802.2** for Logical link control.
 - **IEEE 802.1** for Bridging and management
 - **IEEE 802.10** for security.
- ✓ Later, standards starting from 802.3 are defined such as:
 - **Ethernet** - IEEE 802.3
 - **WiFi** - IEEE 802.11 follows the standards set by 802.1 and 802.2

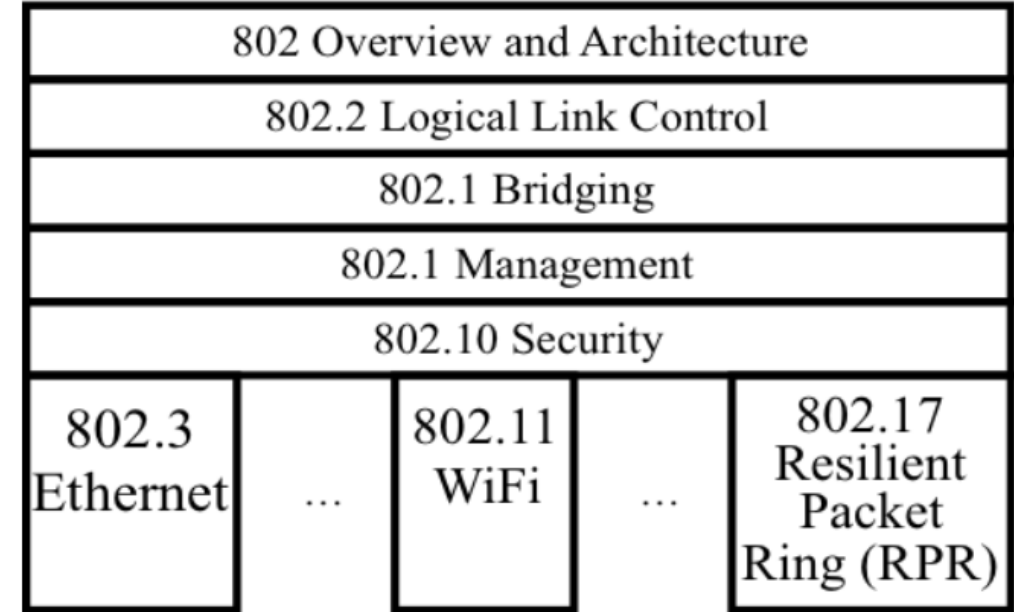


Fig. 2. Hierarchy of IEEE 802 numbering system.

All networks from 802.3 onwards adhere to the same link control, bridging, management, and security standards.

Standards Numbering System

- ✓ Appended letters after an 802 standard apply only to that specific network
 - **Example : 802.11i for WiFi, not Ethernet 802.3.**
- ✓ **Lower case** letters are amendments and will disappear after merging with a base standard.
- ✓ **Upper case** letters represent permanent base standards.
 - **Example: IEEE 802.1w-2001 (amendment) merged with IEEE 802.1D-2004 (base standard).**
- ✓ Standards were initially numbered sequentially (802.1a to 802.1z, then 802.1aa, 802.1ab, etc.).
- ✓ Now, base standard letters are shown during amendments
 - **Example : IEEE 802.1Qau-2010, where Q is the base standard and au is the amendment.**

Standards Numbering System

- ❖ IEEE uses letters for different technology versions, while the WiFi Alliance uses numbers.
- ❖ WiFi version numbers: **WiFi 4** = **IEEE 802.11n**, **WiFi 5** = **802.11ac**, etc.
- ❖ Table 1 shows WiFi Alliance numbers and their corresponding IEEE standards.

WiFi Alliance numbers and their corresponding IEEE standards.

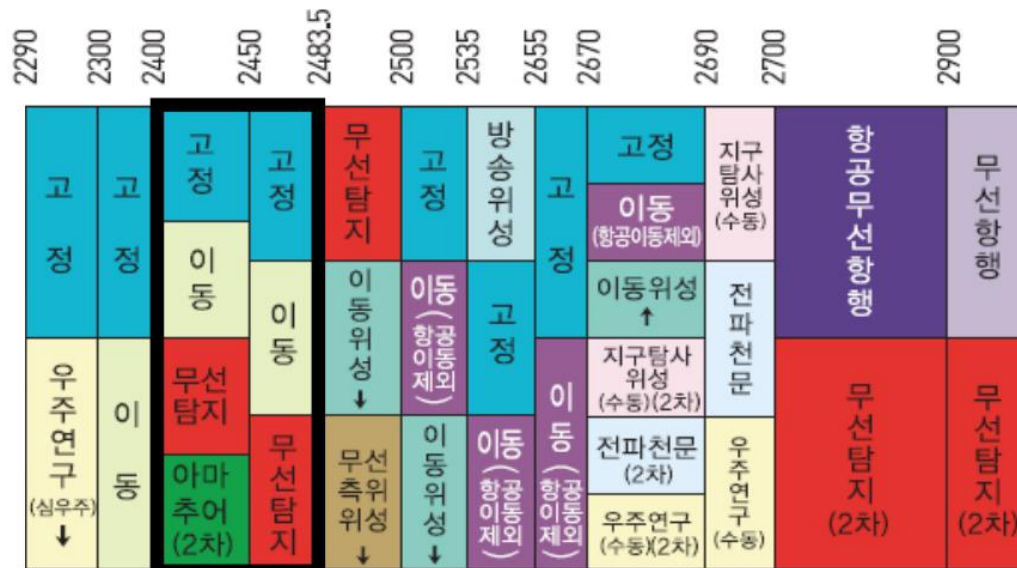
WiFi Alliance Number	IEEE Standard
WiFi 4	802.11n
WiFi 5	802.11ac
WiFi 6	802.11ax
WiFi 7	802.11be

IEEE 802.11 Features

- ✓ The original 802.11 standard (1997) specified speeds of **1** and **2** Mbps.
- ✓ **Newer versions** offered higher speeds:
 - **11 Mbps, 54 Mbps, 108 Mbps, 200 Mbps, etc.**
- ✓ These versions operate on 'license-free' spectrum.
- ✓ **To prevent** interference in license-free bands:
 - **IEEE802.11 uses spread spectrum techniques at the physical layer.**
 - **The ISM (Industrial, Scientific, and Medical) bands**, 802.11 employs:
 - Direct Sequence (DS) spread spectrum
 - Frequency Hopping spread spectrum.
- ✓ **Support priority:** time-critical (e.g., voice) and data traffic on the same LAN.
- ✓ **Power management:** allowing nodes to enter sleep mode when **idle** to conserve power.

Unlicensed Spectrums

- ☑ Can be used without license as long as the regulatory requirements are met
 - ✓ Max transmit power level, spectrum mask, duty cycle, max tx time, etc.
- ☑ Availability depends on countries:\
 - ✓ 2.4 GHz: Wi-Fi, Bluetooth, ZigBee, cordless telephone, and microwave oven.
 - ✓ 5 GHz: Wi-Fi, radars, (and potentially, LTE in unlicensed spectrum)



2.4GHz



5GHz

Industrial, Scientific, and Medical (ISM) Bands

☑ The license-free spectrum for wireless LANs is called the **I**ndustrial, **S**cientific, and **M**edical (**ISM**) band.

☑ ISM bands vary in bandwidth:

- Lower frequency bands having smaller bandwidths
- Higher frequency bands having larger bandwidths.
- Examples:
 - ✓ 6.765 MHz band has a 30 kHz bandwidth,
 - ✓ 5.725 GHz band has a 150 MHz bandwidth.

Table 2. ISM bands.

Frequency Range	Bandwidth
6.765 MHz–6.795 MHz	30 kHz
13.553 MHz–13.567 MHz	14 kHz
26.957 MHz–27.283 MHz	326 kHz
40.660 MHz–40.700 MHz	40 kHz
433.050 MHz–434.790 MHz	1.74 MHz
902 MHz–928 MHz	26 MHz
2.4 GHz–2.5 GHz	100 MHz
5.725 GHz–5.875 GHz	150 MHz
24.000 GHz–24.250 GHz	250 MHz
61 GHz–61.5 GHz	500 MHz
122 GHz–123 GHz	1 GHz
244 GHz–246 GHz	2 GHz

Industrial, Scientific, and Medical (ISM) Bands

- ✓ **The original** WiFi started using the 2.4 GHz band and used by various products:
 - Examples: medical equipment, microwaves, garage-door openers.
- ✓ **Dual band** :WiFi routers can operate on both 2.4 GHz and 5 GHz bands.
- ✓ **Bands** have been released in recent years to support new types of WiFi.

Table 3. WiFi bands.

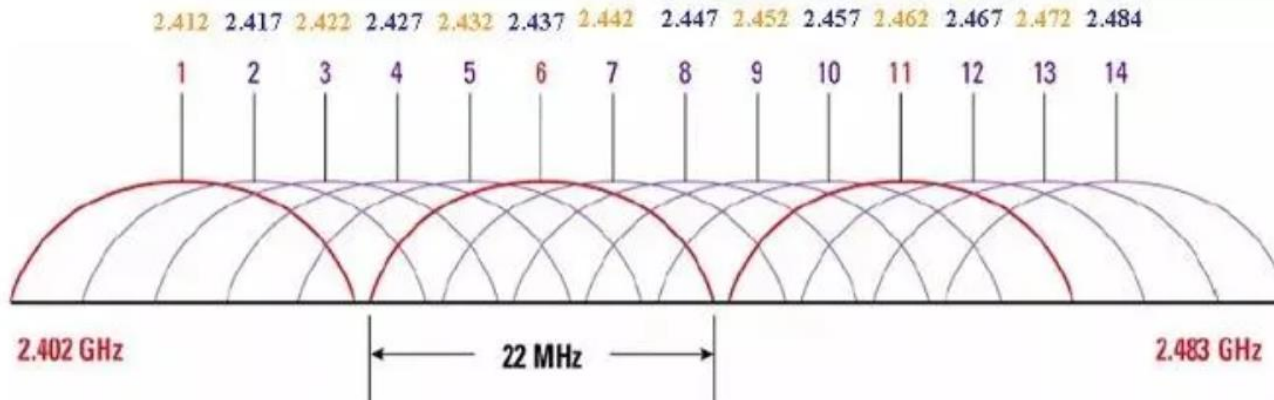
WiFi Standard	Frequency Band
802.11b/g/n	2.4 GHz
802.11a/n/ac/ax	5 GHz
802.11be	6 GHz (not confirmed yet)
802.11p (car-to-car)	5.9 GHz (licensed band)
802.11ah (IoT)	900 MHz
802.11af (Rural)	700 MHz (unused TV channels)
802.11ad/ay (Multi Gbps wireless applications: e.g., cable replacement, VR, ...)	60 GHz

IEEE 802.11 Channels

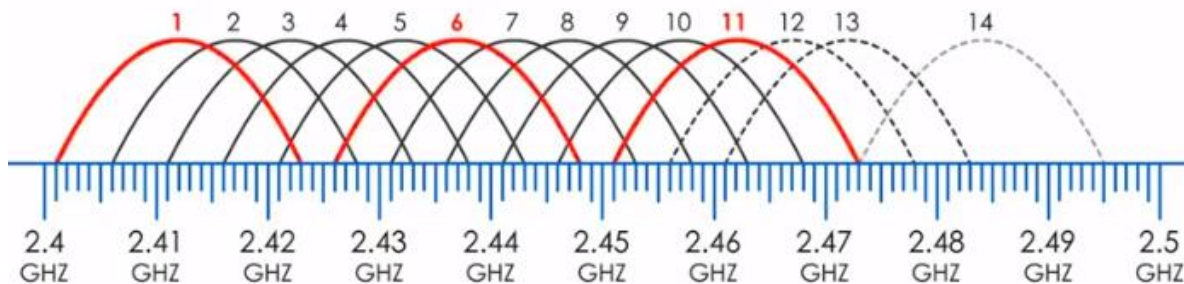
- ☑ WiFi bands are divided into channels to manage **congestion** when multiple wireless LANs operate nearby.
- ☑ Different LANs can operate on different channels within the same band to avoid **collisions** and **interference**.
- ☑ An Access Point (AP) and its connected devices operate on **a single channel** selected by the AP.
- ☑ Channels are usually 20 or 22 MHz wide:
 - *2.4 GHz WiFi uses 22 MHz channels*
 - *5 GHz WiFi uses 20 MHz channels.*
- ☑ Newer WiFi versions can **combine multiple channels** for higher data rates.

IEEE 802.11 Channels: Channels @ 2.4GHz for 802.11b/g/n

- ❖ The 2.4 GHz band has 14 channels, but not all are available in every country.
- ❖ Each 2.4 GHz channel is 22 MHz with center frequencies 5 MHz apart, except for channel 14.

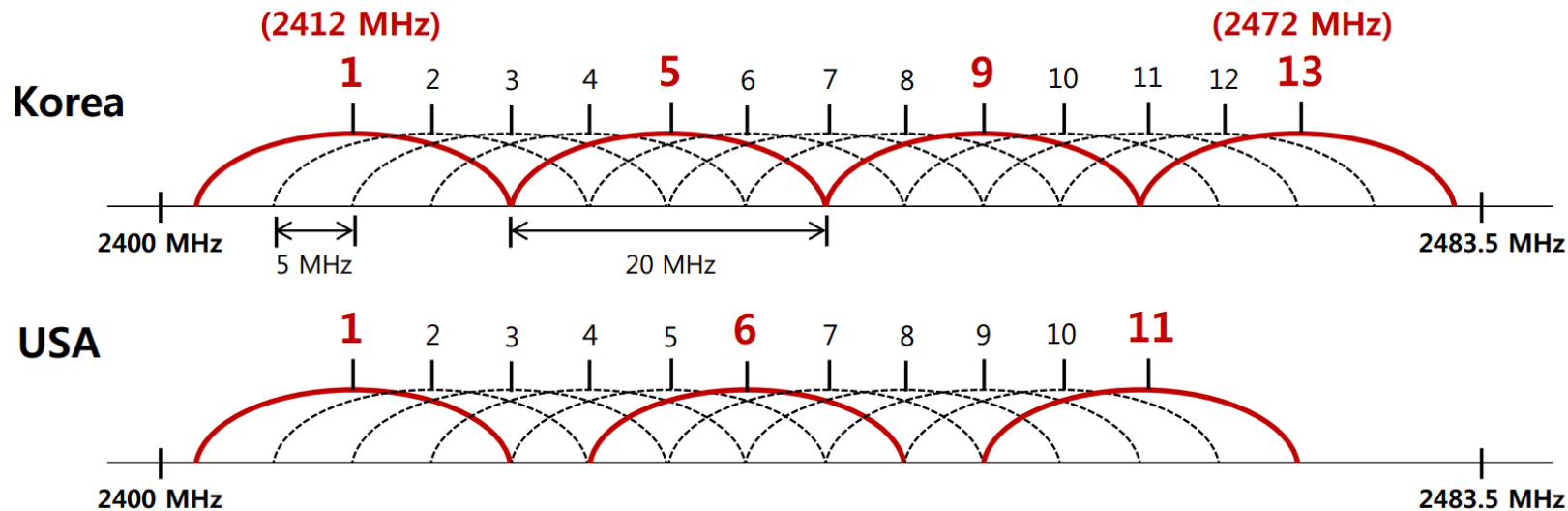


- ❖ Only three non-overlapping channels (1, 6, 11) are available in the 2.4 GHz band.

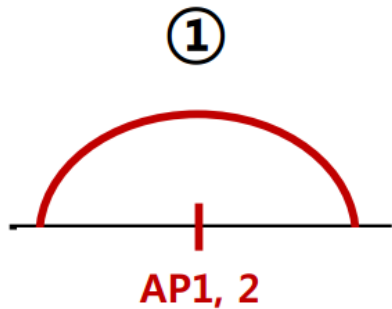


- ❖ Most WiFi routers default to channel 6

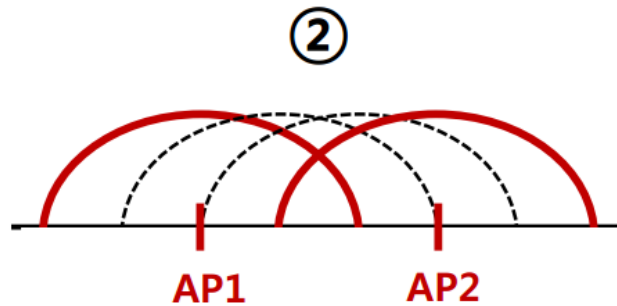
IEEE 802.11 Channels: Channels @ 2.4GHz for 802.11b/g/n



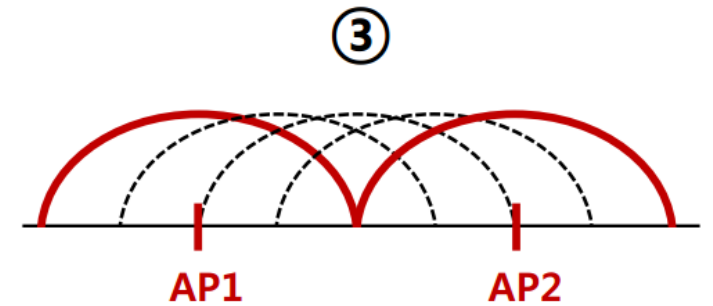
□ Channel use cases in two-cell network



Same channel (fully overlapping)



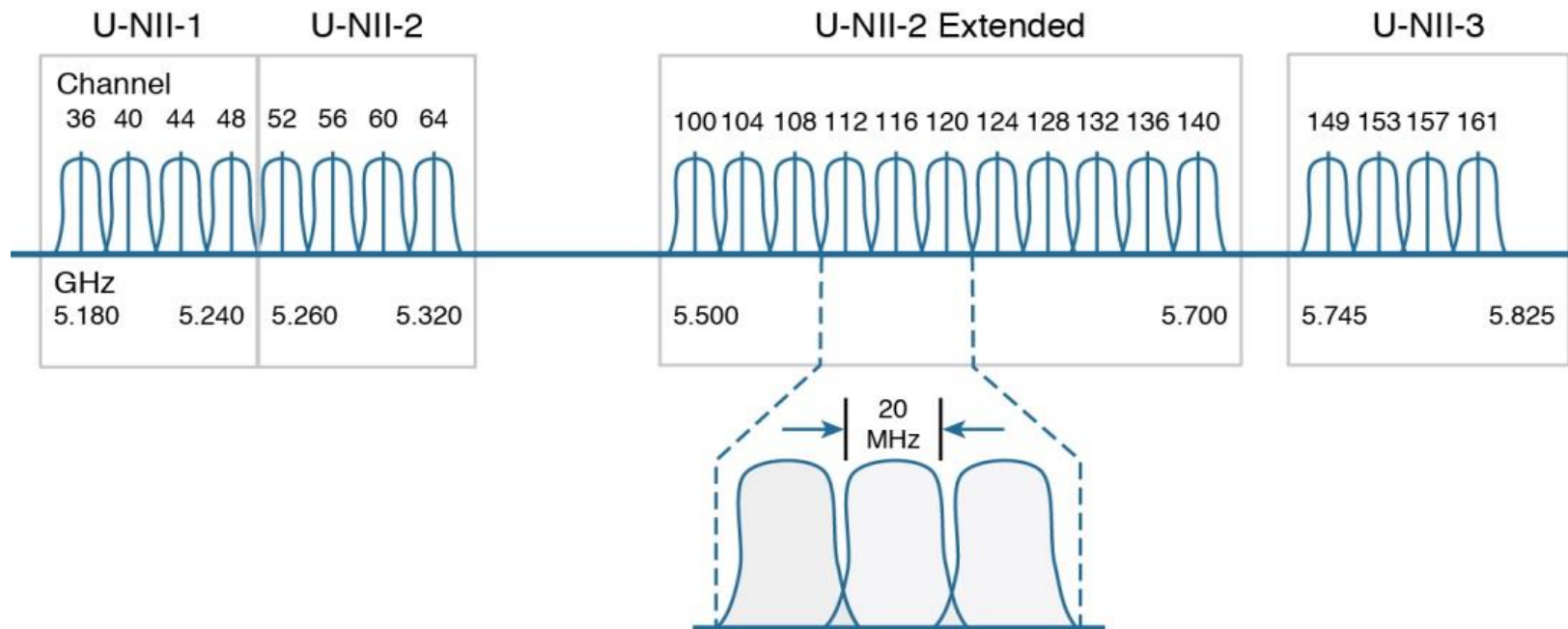
Adjacent channels (partially overlapping)



Orthogonal channels (non-overlapping)

IEEE 802.11 Channels: Channels @ 5GHz for 802.11a/n

- ❖ The 5 GHz band has 20 MHz non-overlapping channels.
- ❖ 5 GHz channels include always-available channels and shared radar channels using **D**ynamic **F**requency **S**election (**DFS**).
- ❖ With DFS, 5 GHz WiFi APs switch channels if radar is detected, causing connected devices to switch as well, which may result in momentary (temporary) connection drops (disconnected).



IEEE 802.11 Channels: Channels @ 5GHz for 802.11a/n

Table 4. Channel frequencies for 2.4 GHz WiFi.

Channel Number	Lower Frequency (MHz)	Center Frequency (MHz)	Upper Frequency (MHz)
1	2401	2412	2423
2	2406	2417	2428
3	2411	2422	2433
4	2416	2427	2438
5	2421	2432	2443
6	2426	2437	2448
7	2431	2442	2453
8	2436	2447	2458
9	2441	2452	2463
10	2446	2457	2468
11	2451	2462	2473
12	2456	2467	2478
13	2461	2472	2483
14	2473	2484	2495

Spaced 5 MHz

Physical Layers

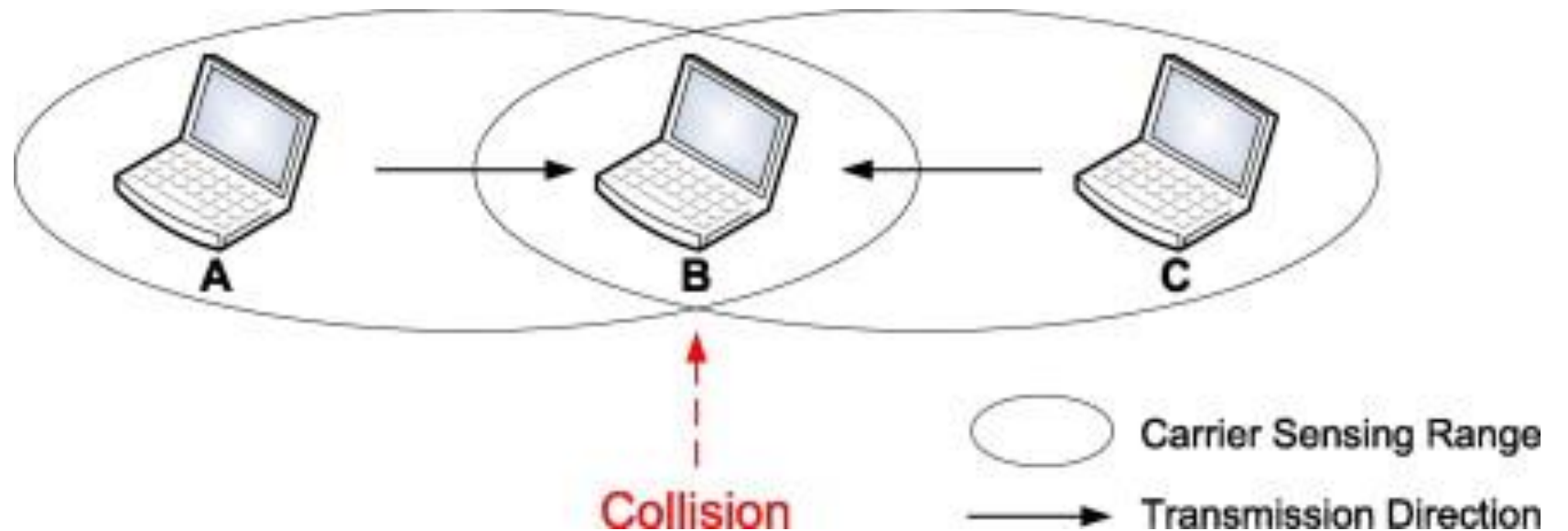
- ❖ Physical layer technology impacts the data rate achievable with WiFi.
 - In 1997, spread spectrum was used to achieve 1 and 2 Mbps.
 - In 1999, an advanced version of spread spectrum was introduced for 2.4 GHz.
 - In 1999, OFDM was introduced for the 5 GHz band, increasing the data rate to 54 Mbps.
 - In 2003, OFDM was used in the 2.4 GHz band, achieving 54 Mbps channels, known as IEEE802.11g.
 - OFDM continues to define the physical layer standard for modern WiFi.

Hidden Node Problem

✓ **Hidden Node Problem:** A specific collision issue in wireless LANs where not all nodes can hear each other.

✓ **Example Scenario:** Nodes A, B, and C:

- A can hear B.
- B can hear C.
- C cannot hear A.



✓ **Collision Example:**

- C starts transmitting to B while A is also transmitting to B.
- B experiences collisions but cannot distinguish between A's and C's transmissions.

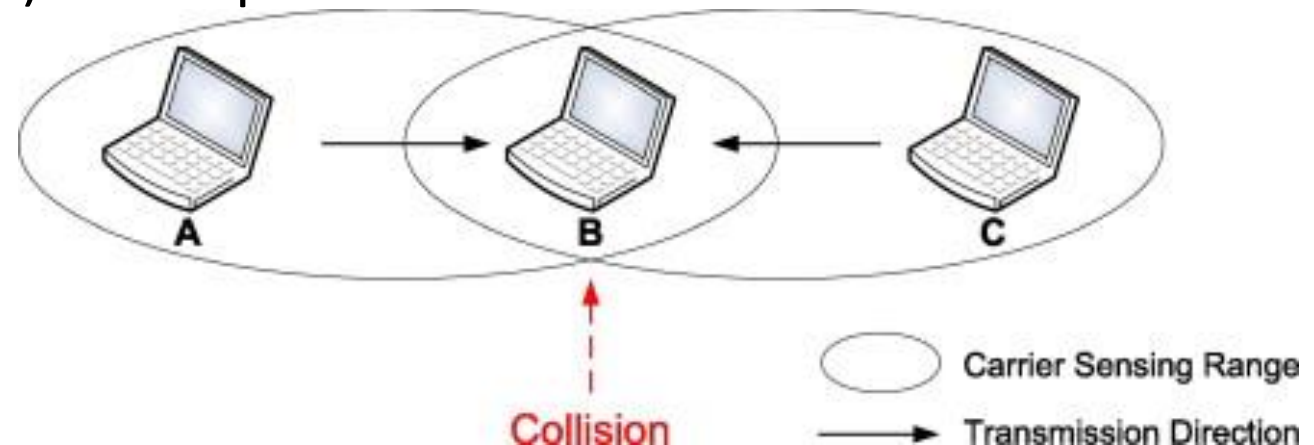
Hidden Node Problem

☑ Collision Detection Limitation:

- In Ethernet, transmitters detect collisions and back off, in wireless LANs:
- In wireless: not all transmitters can detect collisions: like transmitters A and C cannot detect collisions, only the receiver (B) detects collisions.

☑ To solve collision problem:

- Ethernet uses **c**ollision **d**etection (**CD**) technique.
- Wireless uses **c**ollision **a**voidance (**CA**) technique.



Hidden Node Problem

Collision Avoidance with 4-way Handshake

❖ **Purpose:** Achieves CA by using four packets for **each data packet**, with three control packets needed to avoid collisions.

❖ **Process:**

- 1) **RTS (Ready-to-Send):** The transmitter sends an RTS packet to request permission to send data.
- 2) **CTS (Clear-to-Send):** The receiver responds with a CTS packet if it allows the transmission.
- 3) **Data Packet:** The transmitter sends the data packet only after receiving the CTS, ensuring other potential transmitters defer their transmissions.
- 4) **Acknowledgment (ACK):** The receiver sends an ACK to confirm successful reception of the data packet.

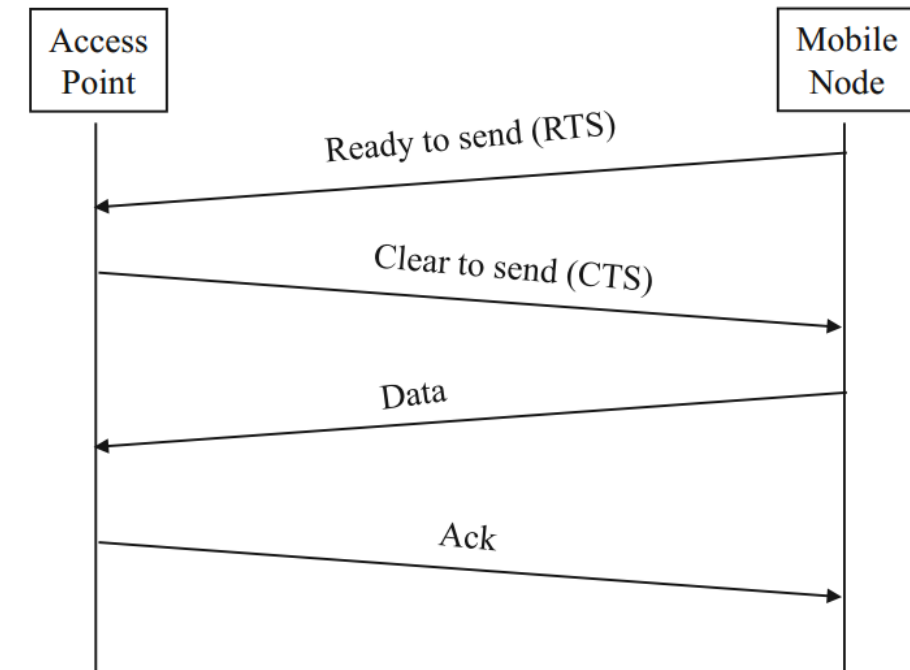


Fig. 4. 4-way handshake using RTS/CTS to avoid collision in 802.11.

Hidden Node Problem

Collision Avoidance with 4-way Handshake

- ❖ **Repetition:** This procedure is repeated for each data packet.
- ❖ **Overhead Consideration:**
 - **RTS/CTS Packets:** Very short (lasting a few microseconds).
 - **Data Packets:** Much longer (tens to hundreds of milliseconds).
 - **Efficiency:** The overhead of RTS/CTS is justified to avoid collisions with long data packets, making the process worthwhile.

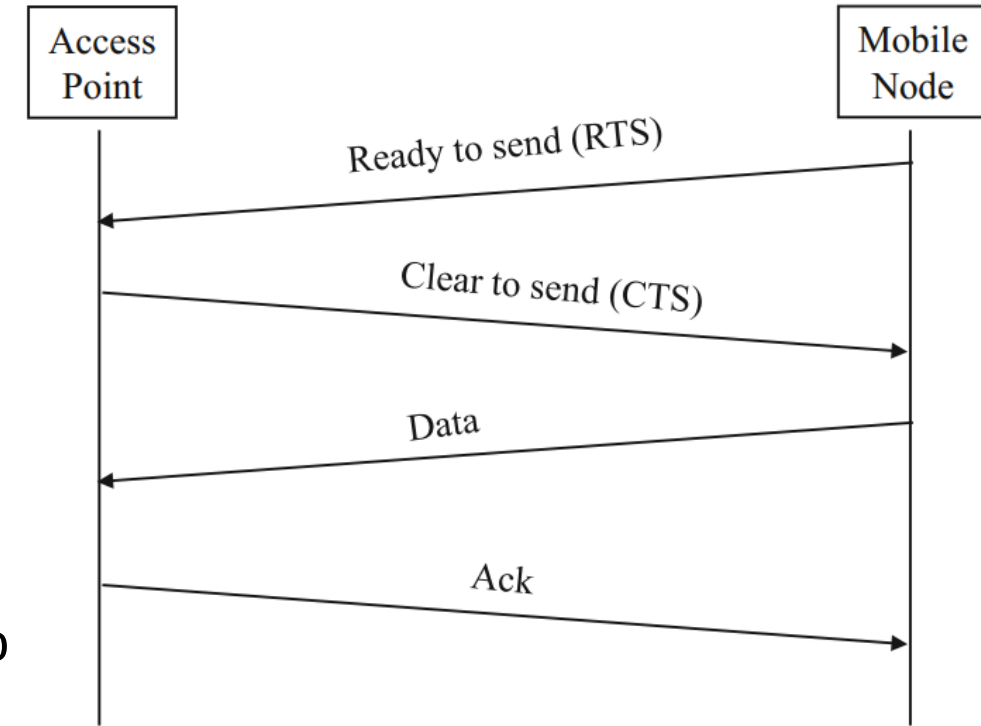


Fig. 4. 4-way handshake using RTS/CTS to avoid collision in 802.11.

THANKS

